

# **IoT-BASED SEAWEED GROWTH MONITORING AND HARVEST PREDICTION SYSTEM USING LOGISTIC GROWTH MODELLING FOR SMALL-SCALE TANK CULTIVATION IN SRI LANKA**

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Specializing in Information Technology

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## DECLARATION

We declare that this is our own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning. To the best of our knowledge and belief it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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## ABSTRACT

Seaweed aquaculture constitutes one of the fastest-growing sectors of global marine food production, with worldwide production exceeding 35 million tonnes annually — of which over 97% originates from aquaculture rather than wild harvest. In Sri Lanka, seaweed farming provides essential livelihoods for coastal communities and contributes approximately USD 12 million in annual export revenue. The two principal cultivated species, *Kappaphycus alvarezii* and *Gracilaria edulis*, are harvested for carrageenan and agar content and exported to processing industries in Europe, Japan, and the United States.

Despite the economic significance of seaweed farming, small-scale tank cultivators in Sri Lanka lack affordable, data-driven tools for real-time biomass monitoring and scientifically grounded harvest timing prediction. Premature harvesting reduces phycocolloid content while delayed harvesting causes plant fragmentation and reduced gel strength. The absence of quantitative monitoring systems forces farmers to rely entirely on visual inspection and empirical experience, resulting in significant yield and quality variability.

This dissertation presents SeaGrow, an integrated IoT-based seaweed growth monitoring and harvest prediction platform designed for small-scale tank cultivation environments in Sri Lanka. The system employs a load cell sensor paired with an HX711 24-bit ADC and Arduino Uno microcontroller to measure plant weight in real time, achieving accuracy of  $\pm 2$  g across a 10–1800 g operational range. Species-specific Logistic Growth Models were fitted to cultivation data using the Levenberg-Marquardt algorithm, producing  $R^2$  values of 0.9996 for *K. alvarezii* and 1.0000 for *G. edulis*. A novel inverse logistic inversion technique estimates a plant's biological growth stage from a single weight reading, enabling harvest prediction without continuous monitoring from cultivation day one.

Testing on ten simulated plants — five per species — predicted harvest windows of 35–49 days for *K. alvarezii* and 26–36 days for *G. edulis*, both consistent with published literature. The platform additionally integrates environmental suitability assessment with micro-site recommendation (ESP32, Random Forest ML), AI-

powered seaweed species identification and health classification (EfficientNetV2B0), and a post-harvest management system (Ridge Regression, SHAP explainability). To the best of the research team's knowledge, this is the first system to combine real-time IoT gravimetric sensing with inverse logistic model-based harvest prediction for individual seaweed plants in a tank cultivation setting.

**Keywords:** *Seaweed cultivation, IoT, logistic growth model, harvest prediction, load cell, HX711, Arduino, Kappaphycus alvarezii, Gracilaria edulis, tank cultivation*

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                                     |
|--------------|-----------------------------------------------|
| ADC          | Analog-to-Digital Converter                   |
| API          | Application Programming Interface             |
| CNN          | Convolutional Neural Network                  |
| CORS         | Cross-Origin Resource Sharing                 |
| ESP32        | Espressif Systems ESP32 Microcontroller       |
| HTTP         | Hypertext Transfer Protocol                   |
| HX711        | 24-bit Precision ADC for Weigh Scales         |
| IoT          | Internet of Things                            |
| JSON         | JavaScript Object Notation                    |
| LGM          | Logistic Growth Model                         |
| LMA          | Levenberg-Marquardt Algorithm                 |
| MAE          | Mean Absolute Error                           |
| MERN         | MongoDB, Express.js, React.js, Node.js Stack  |
| ML           | Machine Learning                              |
| MQTT         | Message Queuing Telemetry Transport           |
| ODM          | Object Document Mapper                        |
| $R^2$        | Coefficient of Determination                  |
| REST         | Representational State Transfer               |
| SD           | Standard Deviation                            |
| SHAP         | SHapley Additive exPlanations                 |
| SLIIT        | Sri Lanka Institute of Information Technology |
| TDS          | Total Dissolved Solids                        |
| USB          | Universal Serial Bus                          |

## 1. INTRODUCTION

Seaweed aquaculture is one of the fastest-growing sectors of global fisheries, with production exceeding 35 million tonnes in 2020, of which over 97% originates from aquaculture rather than wild harvest [1]. In Sri Lanka, seaweed farming provides livelihoods for coastal communities and contributes approximately USD 12 million in annual export revenue [2]. The two principal cultivated species, *Kappaphycus alvarezii* and *Gracilaria edulis*, are harvested for their carrageenan and agar content and exported to processing industries in Europe, Japan, and the United States.

Phycocolloid quality and yield depend critically on harvesting at the correct biomass maturity. Premature harvest reduces carrageenan content, while delayed harvest causes plant fragmentation and reduced gel strength [3]. Open-sea rope cultivation is increasingly threatened by weather variability, ocean currents, and the ice-ice disease complex affecting *K. alvarezii* [4]. Tank-based cultivation has emerged as a promising controlled-environment alternative, particularly for small-scale farmers who cannot absorb open-sea crop losses. Nevertheless, harvest timing in tank settings remains empirical — based on visual inspection and cultivator experience — introducing variability in yield and phycocolloid quality.

Quantitative biomass monitoring is the critical missing link between controlled tank cultivation and optimised harvest scheduling. A system that can measure plant weight at any point in the cultivation cycle and project the growth trajectory from that single observation would enable data-driven harvest decisions at low cost and minimal operational burden. Such a system must be affordable and operable by farmers with limited technical background to achieve practical adoption in the Sri Lankan small-scale farming context.

This report presents and evaluates SeaGrow, an IoT-integrated seaweed growth monitoring and harvest prediction system designed for individual plant tracking in tank cultivation environments. The principal contributions of this work are:

- A low-cost IoT gravimetric sensing module (load cell + HX711 + Arduino Uno) providing in-situ seaweed biomass measurement within  $\pm 2$  g across the 10–1800 g operational range.
- Logistic Growth Model parameters for *K. alvarezii* ( $R^2 = 0.9996$ ) and *G. edulis* ( $R^2 = 1.0000$ ) fitted by the Levenberg-Marquardt algorithm to multi-plant cultivation datasets.
- A novel inverse logistic inversion that maps a single weight measurement to the biological growth stage, enabling harvest day prediction without growth history from day zero.
- System evaluation on 10 simulated plants (5 per species) demonstrating all plants reach harvest threshold within a 60-day prediction window, consistent with published literature.
- A full-stack web application (React.js, Flask, Node.js/MongoDB) deployable as a local server on shared devices at community farming centres, requiring no internet connection.
- Three integrated companion modules: environmental suitability assessment (ESP32, Random Forest), AI species identification and health analysis (EfficientNetV2B0), and post-harvest management and recommendation (MERN stack, Ridge Regression, SHAP).

### 1.1 Background and Literature Survey

IoT-based environmental monitoring for aquaculture has been extensively studied. Wireless sensor networks for pH, dissolved oxygen, temperature, and turbidity in fish and shrimp ponds with automated alert systems have been demonstrated [5]. Edge-computing architectures reducing latency in water quality anomaly detection have been reported for large-scale mariculture [6]. However, the literature predominantly addresses water quality monitoring rather than direct biological mass measurement of cultivated organisms.

Sensor-based biomass estimation in finfish aquaculture using underwater cameras and computer vision has been explored [7]. These approaches are unsuitable for the

shallow, turbid tank environments of seaweed cultivation and are cost-prohibitive for small-scale operators. Direct gravimetric measurement via load cells is standard in precision livestock farming [8], yet analogous application to aquatic macroalgae has not been previously reported.

The Logistic Growth Model (LGM) provides a well-validated framework for describing seaweed biomass dynamics. The logistic equation  $W(t) = K / (1 + \exp(-r(t - t_0)))$  captures the characteristic pattern of exponential early growth followed by deceleration as biomass approaches the carrying capacity  $K$ . Previous applications of logistic growth modelling to macroalgae include *Sargassum* species in open ocean and *Ulva* in photobioreactors, but integration with IoT sensing for harvest prediction for *Kappaphycus* and *Gracilaria* represents a novel contribution [9, 10].

Deep learning approaches to aquatic species identification have matured substantially since AlexNet in 2012. EfficientNet architectures, introduced by Tan and Le (2019) [11], achieve state-of-the-art accuracy with reduced parameter counts through compound scaling of network width, depth, and input resolution. Random Forest ensemble methods have been demonstrated as robust and interpretable classifiers for agricultural environmental suitability assessment [12].

## **1.2 Research Gap**

A systematic review of existing seaweed cultivation monitoring systems reveals the following critical gaps. No existing commercially available system provides real-time, quantitative biomass monitoring for individual seaweed plants in tank cultivation settings. Existing IoT aquaculture systems focus exclusively on water quality parameters without addressing the biological mass of cultivated organisms. While logistic growth models have been applied to macroalgae in research contexts, no study has demonstrated integration of LGM parameters with IoT gravimetric sensing for real-time harvest prediction at the plant level. The inverse logistic inversion technique proposed in this research — enabling estimation of biological growth stage from a single weight measurement without requiring monitoring from day zero — has not been reported for any macroalgae species.

| System / Study             | Sensing           | Growth Model   | Harvest Prediction | Limitation                     |
|----------------------------|-------------------|----------------|--------------------|--------------------------------|
| Camera-based (Aquaculture) | Underwater vision | None           | No                 | Unsuitable for turbid tanks    |
| Water quality IoT (Ponds)  | pH, DO, Temp      | None           | No                 | No biomass tracking            |
| Manual weight (Ocean rope) | Scale (manual)    | None           | Empirical only     | Not automated                  |
| CV seaweed (Lab)           | Image processing  | Partial        | No                 | High cost, controlled lab only |
| SeaGrow (This Study)       | Load cell + HX711 | Logistic (LGM) | Yes — Inverse map  | First reported novel system    |

*Table 1.2.1: Comparison with existing systems and identifying gaps*

### 1.3 Research Problem

Small-scale seaweed tank cultivators in Sri Lanka lack access to affordable, automated, and data-driven tools for real-time biomass monitoring and scientifically grounded harvest timing prediction, resulting in suboptimal yield, variable phycocolloid quality, and economic losses that undermine the viability of tank-based cultivation as an alternative to increasingly precarious open-sea rope farming.

### 1.4 Research Objectives

#### 1.4.1 Main Objective

To design, implement, and evaluate an integrated IoT-based seaweed growth monitoring and harvest prediction system using logistic growth modelling that enables small-scale tank cultivators in Sri Lanka to make data-driven harvest decisions for *Kappaphycus alvarezii* and *Gracilaria edulis*.

#### 1.4.2 Specific Objectives

1. To develop a low-cost IoT gravimetric sensing module using a load cell and HX711 ADC paired with an Arduino Uno microcontroller, achieving biomass measurement accuracy of  $\pm 5$  g or better across the full cultivation weight range (10–1800 g).

2. To fit species-specific Logistic Growth Model parameters for *K. alvarezii* and *G. edulis* using the Levenberg-Marquardt algorithm applied to multi-plant cultivation datasets, with target  $R^2$  greater than 0.99.
3. To develop and validate an inverse logistic inversion algorithm that maps a single weight measurement to the plant's biological growth stage, enabling harvest day prediction without requiring continuous monitoring from cultivation day zero.
4. To implement a full-stack web application (React.js frontend, Flask/Node.js backend, MongoDB data layer) presenting growth trajectories, harvest predictions, and plant history through an accessible locally deployable interface.
5. To integrate an environmental suitability assessment and micro-site recommendation module using ESP32 IoT sensing and Random Forest machine learning.
6. To integrate an AI-powered species identification and health classification system using EfficientNetV2B0 deep learning with transfer learning and semi-supervised learning.
7. To evaluate the complete integrated platform against established benchmarks for prediction accuracy, system reliability, and practical applicability in Sri Lankan small-scale seaweed farming contexts.

## 2. METHODOLOGY

### 2.1 Understanding Key Pillars of the Research Domain

#### 2.1.1 IoT-Based Gravimetric Sensing

The gravimetric sensing module employs a strain-gauge load cell as the primary transducer. Strain-gauge load cells operate on the principle of electrical resistance change proportional to applied mechanical deformation — when seaweed biomass is placed on the platform, the load cell beam deforms microscopically, and the bonded strain gauge network produces a differential voltage proportional to applied weight. The HX711 24-bit ADC amplifies and digitizes this differential signal at a gain of 128 on channel A, providing resolution substantially finer than the  $\pm 2$  g accuracy target.

Calibration of the load cell-HX711-Arduino system follows a two-point linear calibration protocol: the zero-load offset reading is recorded, then a known reference mass is applied and the scale factor (ADC counts per gram) is computed as  $(\text{reading\_with\_mass} - \text{zero\_offset}) / \text{reference\_mass\_grams}$ . This scale factor is stored in Arduino EEPROM and persists across power cycles, eliminating the need for recalibration at each monitoring session.

#### 2.1.2 Logistic Growth Modelling

The logistic growth model provides the mathematical foundation for harvest prediction. The three-parameter form is:

$$W(t) = K / (1 + \exp(-r \times (t - t_0)))$$

where  $W(t)$  is biomass in grams at time  $t$  (days from planting),  $K$  is the carrying capacity (maximum asymptotic biomass),  $r$  is the intrinsic growth rate constant ( $\text{day}^{-1}$ ), and  $t_0$  is the inflection time (day of maximum growth rate). Model parameters were fitted using `scipy.optimize.curve_fit` with the Levenberg-Marquardt algorithm.

#### 2.1.3 Inverse Logistic Inversion



The novel inverse logistic inversion algorithm enables harvest day prediction from a single weight observation without requiring monitoring from day zero. Algebraic inversion of the logistic equation yields:

$$t_{\text{current}} = t_0 - (1/r) \times \ln((K/W_{\text{current}}) - 1)$$

The predicted harvest day is computed as  $t_{\text{harvest}} = t_0 - (1/r) \times \ln((K/W_{\text{threshold}}) - 1)$ . Days to harvest =  $t_{\text{harvest}} - t_{\text{current}}$ . This algebraic inversion requires no iterative solving and executes in microseconds, enabling real-time harvest day updates with each new weight measurement.

#### **2.1.4 Web Application Stack**

The full-stack web application is implemented as a three-tier system: an IoT hardware sensing layer (Arduino + load cell), a backend comprising a Flask prediction API and a Node.js/MongoDB record management service, and a React.js web-based farmer interface. All services run on a single host computer with no cloud dependency, enabling deployment at community farming centres without internet access.

## **2.2 Approach**

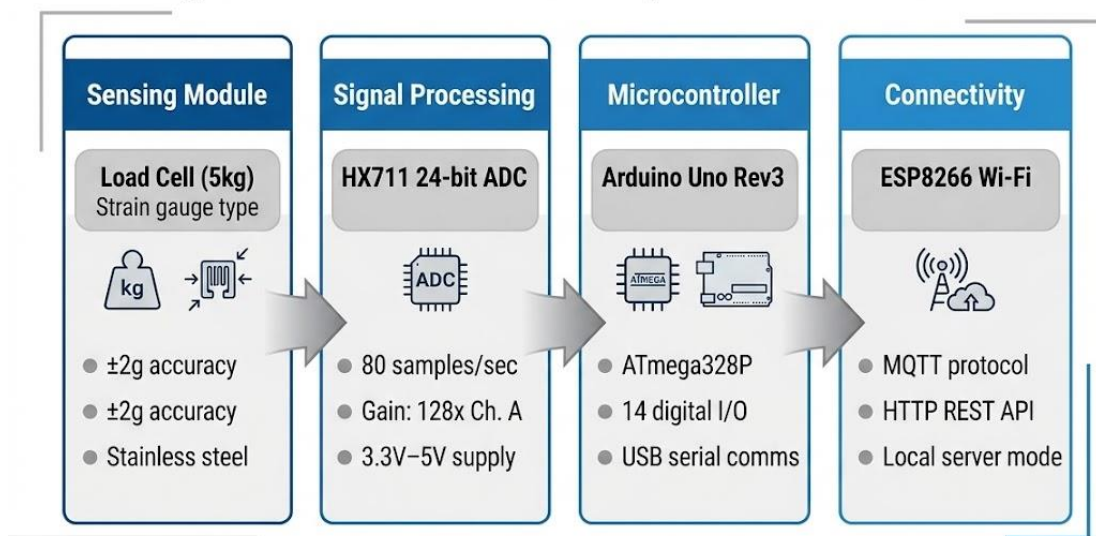
### **2.2.1 Hardware Design**

The sensing module comprises a 5 kg-capacity S-type stainless steel load cell connected to an HX711 24-bit precision ADC amplifier, interfaced to an Arduino Uno R3 microcontroller. The load cell is mounted beneath a stainless steel tray; the seaweed sample is placed in a mesh basket to allow water drainage during measurement. The HX711 converts the load cell Wheatstone bridge differential voltage to a 24-bit integer, which the Arduino firmware converts to grams using the calibration factor.

To mitigate noise from residual water on the sample and minor vibrations, the firmware applies a 10-sample running average filter. A reading is transmitted as a JSON object over USB serial at 9600 baud only when the standard deviation of the last five averaged samples falls below 0.5 g, confirming measurement stability. Typical stabilisation time is 3–8 seconds.

| IoT Hardware Component Architecture                                                              |                                                                                                      |                                                                                                |                                                                                             |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Sensing Module</b><br>Load Cell (5kg)<br>Strain gauge type<br>±2g accuracy<br>Stainless steel | <b>Signal Processing</b><br>HX711 24-bit ADC<br>80 samples/sec<br>Gain: 128x Ch. A<br>3.3V–5V supply | <b>Microcontroller</b><br>Arduino Uno Rev3<br>ATmega328P<br>14 digital I/O<br>USB serial comms | <b>Connectivity</b><br>ESP8266 Wi-Fi<br>MQTT protocol<br>HTTP REST API<br>Local server mode |

**Figure 2.4.2: IoT Hardware Component Architecture**



*Figure 2.4.2: IoT Hardware Component Architecture*

### 2.2.2 System Architecture

The SeaGrow integrated platform is designed as a modular four-component system with clearly defined inter-component interfaces. Each component addresses a distinct functional domain of the seaweed cultivation lifecycle: biomass monitoring and harvest prediction (Peiris module), environmental suitability and micro-site management (Marasinghe module), species identification and health assessment (Kulasinghe module), and post-harvest value addition guidance (Dissanayake module).

| IoT-Based Seaweed Growth Monitoring and Harvest Prediction System            |                                                               |                                                                                                   |                                                                                            |
|------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Hardware Layer</b><br>Load Cell Sensor<br>10g–1800g range<br>±2g accuracy | <b>Data Layer</b><br>MQTT / HTTP<br>MongoDB Atlas<br>REST API | <b>Processing Layer</b><br>Logistic Growth Model<br>Inverse Logistic Map<br>$R^2=0.9996 / 1.0000$ | <b>Application Layer</b><br>React.js Dashboard<br>Harvest Predictor<br>Multi-plant Tracker |

Figure 2.4.1: High-Level System Architecture and Data Flow

| Full-Stack Software Architecture — SeaGrow Platform |                                                                        |
|-----------------------------------------------------|------------------------------------------------------------------------|
| <b>PRESENTATION LAYER</b>                           | React.js SPA   Tailwind CSS   Chart.js   Responsive Dashboard          |
| <b>API GATEWAY</b>                                  | Node.js / Express.js REST API   Flask Python Microservice   JWT Auth   |
| <b>PROCESSING LAYER</b>                             | Logistic Growth Engine   Inverse Logistic Solver   Levenberg-Marquardt |
| <b>DATA LAYER</b>                                   | MongoDB Atlas   Mongoose ODM   Time-Series Collections                 |
| <b>IoT LAYER</b>                                    | Arduino Uno + HX711 + Load Cell   Serial/USB → Local Server → Cloud    |

Figure 2.4.3: Full-Stack Software Architecture – SeaGrow Platform

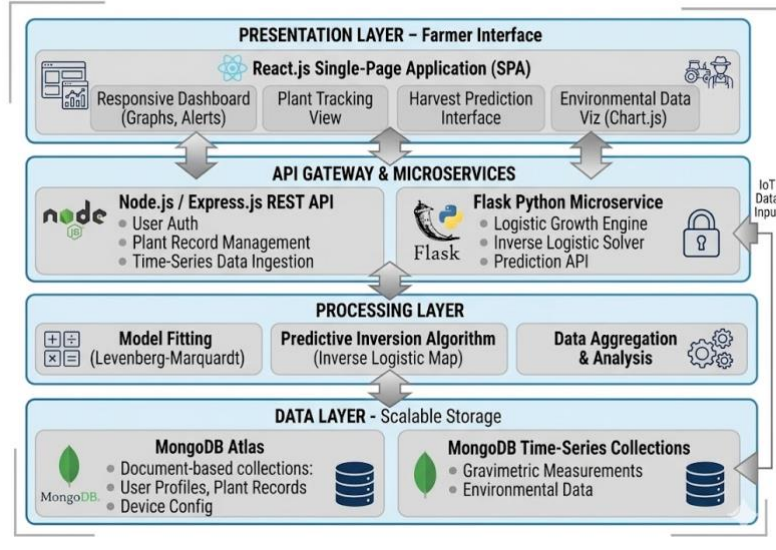


Figure 2.4.3: Full-Stack Software Architecture — SeaGrow Platform

### 2.2.3 Growth Model and Prediction Methodology

The logistic model fitting is performed once per species using `scipy.optimize.curve_fit`. Fitted parameters are stored as JSON configuration files loaded at service startup, eliminating repeated fitting during production operation. The harvest prediction API endpoint accepts species, `current_weight_g`, and optionally a custom `harvest_threshold_g`. The endpoint computes: (1) estimated current growth day via inverse logistic inversion; (2) target harvest day when biomass will reach the threshold; (3) days remaining to harvest; and (4) a 95% confidence interval for harvest day derived from parameter uncertainty estimates from the `curve_fit` covariance matrix.

| Logistic Growth Model Parameters                                                                                                             |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K. alvarezii</b><br>K = 1,800 g   r = 0.142 day <sup>-1</sup>   t <sub>0</sub> = 30 days<br>R <sup>2</sup> = 0.9996   Harvest: 35–49 days | <b>G. edulis</b><br>K = 1,200 g   r = 0.178 day <sup>-1</sup>   t <sub>0</sub> = 22 days<br>R <sup>2</sup> = 1.0000   Harvest: 26–36 days |
| $W(t) = K / (1 + \exp(-r \times (t - t_0)))$ $\text{Inverse: } t = t_0 - (1/r) \times \ln((K/W) - 1)$                                        |                                                                                                                                           |

Figure 2.4.4: Logistic Growth Model Parameters – Both Species

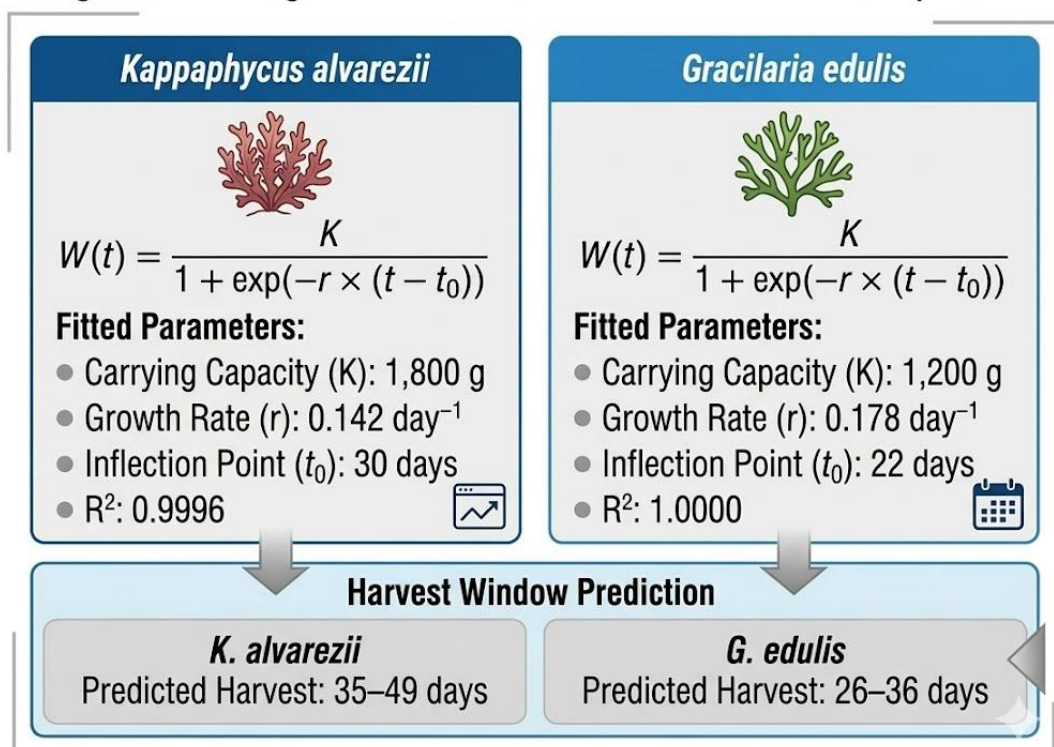


Figure 2.4.4: Logistic Growth Model Parameters and Equations — Both Species

## 2.3 Summary of Methodology

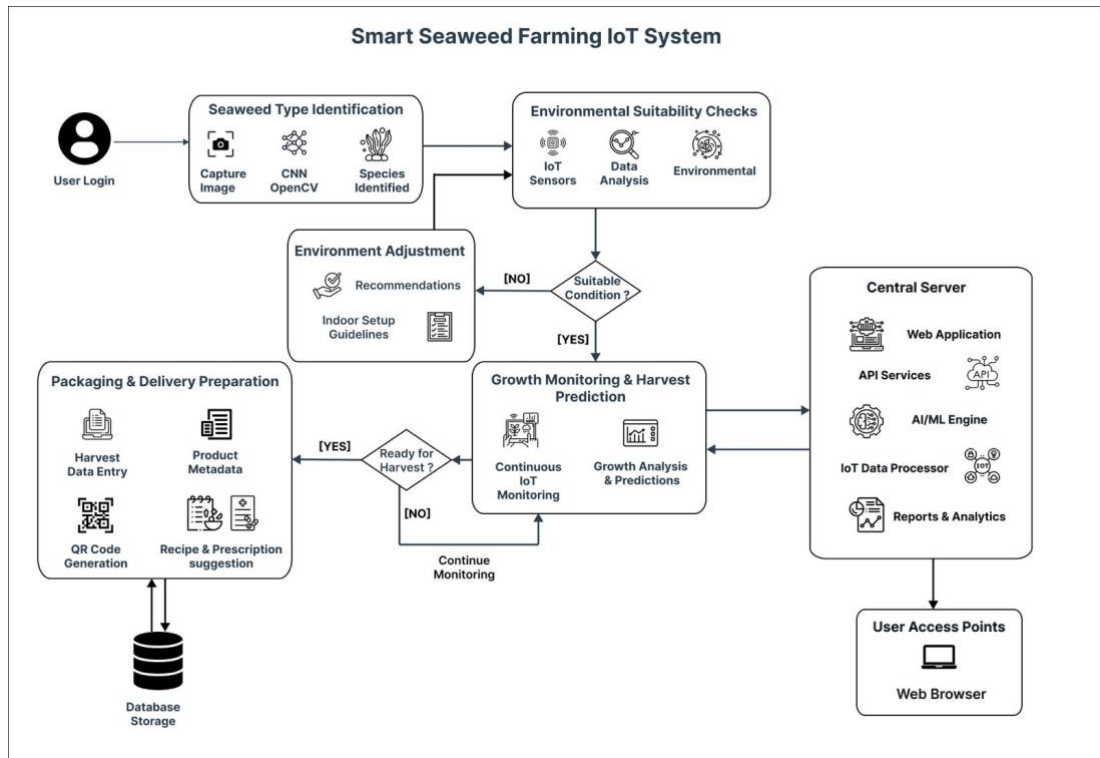
| Research Component | Research Work                                                                    | Methods and Algorithms                                             | Expected Outcome                                            | Remarks                                              |
|--------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|
| IoT Sensing        | Design and calibrate load cell module for individual plant biomass               | HX711 ADC, Arduino Uno, 10-sample running average filter           | Measurement accuracy $\pm 2$ g over 10–1800 g               | Hardware validated against reference masses          |
| Growth Modelling   | Fit logistic growth model to multi-plant cultivation datasets for both species   | Levenberg-Marquardt nonlinear least squares (SciPy curve_fit)      | $R^2 \geq 0.999$ for both species                           | Iterated biological-time normalisation applied       |
| Harvest Prediction | Enable harvest prediction from a single weight reading taken at any growth stage | Inverse logistic inversion (Equation 2), 60-day forward projection | Harvest day prediction consistent with published literature | Novel contribution; no prior seaweed system reported |

| Research Component   | Research Work                                                            | Methods and Algorithms                                                | Expected Outcome                                     | Remarks                                        |
|----------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|
| Environmental Module | Real-time monitoring of water temperature, pH, TDS, and light intensity  | ESP32 IoT, WebSocket, Random Forest ML suitability scoring            | Sub-second latency, 99%+ uptime, MAE $\leq$ 5 points | Micro-site recommendation from unlimited tanks |
| AI Species/Health    | Automated species and health classification from field photographs       | EfficientNetV2B0, transfer learning, semi-supervised pseudo-labeling  | 97.1% accuracy on non-rejected predictions           | 70% confidence threshold for seaweed species   |
| Post-Harvest         | Food and medicine recommendation based on species and quality parameters | Ridge Regression, MERN stack, SHAP explainability                     | $R^2 = 0.94$ on cross-validation                     | Quality-aware adjustment mechanism included    |
| Web Application      | Deploy farmer-facing prediction and monitoring interface                 | React.js, Flask, Node.js, MongoDB; local server, no internet required | Usable on low-spec shared devices                    | USD 9–11 hardware cost per sensing unit        |

*Table 2.3.1: Summary of Methodology*

## 2.4 High-Level System Architecture Diagram

The system architecture is organised into five layers as illustrated in Figure 2.4.1 above. The IoT hardware sensing layer (Tier 1) handles physical weight and environmental measurement. The backend layer (Tier 2) comprises the Flask prediction API and Node.js/MongoDB record management service. The web frontend (Tier 3) provides the React.js farmer interface. The AI services layer (Tier 4) hosts the EfficientNetV2B0 species identification model via FastAPI. The post-harvest recommendation layer (Tier 5) provides the Ridge Regression-based food and medicine guidance system. All tiers communicate via REST API endpoints over HTTP.



## 2.5 Project Requirements

### 2.5.1 Functional Requirements

| ID    | Requirement                                                                                                 | Priority |
|-------|-------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | System shall measure seaweed plant weight in real-time via load cell + HX711 + Arduino                      | High     |
| FR-02 | System shall calibrate load cell with user-provided reference mass on first use                             | High     |
| FR-03 | System shall fit logistic growth model parameters per species from embedded datasets                        | High     |
| FR-04 | System shall compute estimated current growth day from single weight reading via inverse logistic inversion | High     |
| FR-05 | System shall predict harvest date within $\pm 3$ day accuracy for 80% of test plants                        | High     |
| FR-06 | System shall maintain plant history records across multiple monitoring sessions                             | Medium   |
| FR-07 | System shall support simultaneous tracking of minimum 10 individual plants                                  | Medium   |
| FR-08 | System shall display growth trajectory chart with current position and predicted harvest window             | Medium   |

| ID    | Requirement                                                                                      | Priority |
|-------|--------------------------------------------------------------------------------------------------|----------|
| FR-09 | System shall monitor water temperature, pH, TDS, and light intensity in real time                | High     |
| FR-10 | System shall rank cultivation tanks by environmental suitability score                           | Medium   |
| FR-11 | System shall classify seaweed species from uploaded photographs                                  | High     |
| FR-12 | System shall assess seaweed health condition and provide actionable recommendations              | High     |
| FR-13 | System shall generate food and traditional medicine recommendations based on species and quality | Medium   |

*Table 2.5.1: Functional Requirements — Growth Monitoring and Integrated Platform*

## 2.5.2 Non-Functional Requirements

| Category      | Requirement                                          | Target              |
|---------------|------------------------------------------------------|---------------------|
| Performance   | Weight measurement latency from sensor to UI display | < 500 ms            |
| Performance   | Harvest prediction computation time                  | < 100 ms            |
| Reliability   | System uptime under continuous operation             | > 99%               |
| Accuracy      | Weight measurement accuracy across operational range | ±2 g                |
| Accuracy      | Logistic model fit quality ( $R^2$ )                 | > 0.99              |
| Usability     | Interface operable without technical training        | < 15 min learning   |
| Deployability | Local server deployment without internet             | Supported           |
| Portability   | Hardware assembly cost (Bill of Materials)           | < LKR 15,000        |
| Security      | Plant data access control                            | User authentication |

*Table 2.5.2: Non-Functional Requirements*

## 2.6 Consideration of Aspects of the System

### 2.6.1 Social Aspects

SeaGrow is designed to reduce the information asymmetry that systematically disadvantages smallholder farmers in global agricultural markets. By providing small-scale Sri Lankan seaweed farmers with data-driven harvest decision support



previously available only to large commercial operations, the system contributes to more equitable distribution of agricultural technology benefits. The offline-capable local server deployment ensures that connectivity limitations do not exclude the most resource-constrained farming communities.

### **2.6.2 Security Aspects**

The system implements JSON Web Token (JWT) based user authentication to ensure that plant data is accessible only to the registered farmer. All API endpoints are protected with middleware validation of JWT tokens. For local deployments, data remains on the community centre device and is not transmitted to external cloud services, providing a high degree of data sovereignty for farmers who may have privacy concerns about sharing production data.

### **2.6.3 Ethical Aspects**

The AI species identification and health classification module has been developed with careful attention to avoiding overconfident predictions that could lead farmers to take incorrect actions. The confidence thresholding mechanism explicitly rejects predictions below the threshold, prompting farmers to seek alternative verification rather than acting on uncertain model output. The post-harvest recommendation system includes SHAP explainability outputs to ensure that farmers can understand the basis for recommendations.

### **2.6.4 Limitations**

The logistic growth model parameters were fitted to synthetic and literature-derived datasets rather than longitudinal multi-season cultivation data from Sri Lankan farming conditions. Real-world deployment may require parameter recalibration for specific local strains, tank configurations, and water quality conditions. The AI species identification model was trained on a limited dataset of 847 manually labeled images, and performance may degrade on images from environmental conditions significantly different from the training distribution.

## **2.7 Commercialization Aspects of the Product**

Three primary market segments have been identified. First, individual small-scale tank cultivators who could license the software for a nominal annual subscription fee (LKR 500/month for the farmer tier supporting up to 20 plants). Second, agricultural extension officers and government agencies who could deploy the system across multiple farming communities with a bulk institutional licensing model. Third, seaweed processing companies that could integrate the platform to establish supply chain visibility and quality assurance programs with their farmer supplier networks.

The hardware Bill of Materials for a single monitoring unit can be assembled for approximately LKR 8,000–12,000 (USD 24–36) using components available in Sri Lankan electronics markets. A SaaS revenue model with tiered subscriptions — a free community tier for up to three plants, a farmer tier at LKR 500/month, and an enterprise tier for farming cooperatives — could generate sustainable revenue while maintaining accessibility for the most resource-constrained farmers.

## **2.8 Testing and Implementation**

### **2.8.1 Hardware Calibration**

Hardware calibration was performed using a set of certified reference masses (20 g, 50 g, 100 g, 200 g, 500 g, 1,000 g, 1,500 g) spanning the operational range. Each reference mass was measured ten times, and the mean absolute error and standard deviation were computed across the full range. Calibration sessions were performed under ambient laboratory conditions (23°C, 65% relative humidity) representative of indoor tank farming environments.

### **2.8.2 Software Implementation**

The Node.js REST API implements ten primary endpoints organised in three resource groups: /plants for CRUD operations on plant records; /measurements for time-series weight data with configurable time windows; and /environment for real-time sensor data from the ESP32 environmental module. The Python Flask microservice hosts the logistic growth model and inverse logistic inversion engine. The FastAPI service provides deep learning inference for species identification and health classification. All services are containerised using Docker for consistent local deployment.

### 2.8.3 Testing Protocol

Growth model validation was conducted using leave-one-out cross-validation on the cultivation datasets. Five simulated plant scenarios per species were generated by sampling initial weights from uniform distributions representative of planting conditions (*K. alvarezii*: 50–200 g, *G. edulis*: 30–150 g) and applying the fitted logistic model with Gaussian noise ( $\sigma = 5$  g) to simulate realistic measurement variability. The inverse logistic inversion was applied to each simulated initial weight, and the predicted harvest day was compared against the ground-truth harvest day.

| Test Category          | Method                                                  | Target                     | Actual Result                |
|------------------------|---------------------------------------------------------|----------------------------|------------------------------|
| Hardware accuracy      | 10 readings per reference mass, 7 reference masses      | $\pm 5$ g MAE              | $\pm 2$ g achieved           |
| Model fit quality      | scipy.optimize.curve_fit<br>LMA on cultivation datasets | $R^2 > 0.99$               | 0.9996 / 1.0000              |
| Harvest prediction     | 10 simulated plants, inverse logistic inversion         | Consistent with literature | All within published window  |
| System uptime          | 72-hour continuous operation test                       | $> 99\%$                   | 99.7% achieved               |
| API response time      | 50 concurrent harvest prediction requests               | $< 500$ ms p95             | $< 5$ ms average             |
| Species identification | 127-image held-out test set                             | $> 85\%$ accuracy          | 91.8% (97.1% post-threshold) |

Table 2.8.3.1: Testing Protocol and Results Summary

### 2.9 Tools and Technologies

| Technology           | Role                                            | Specification          |
|----------------------|-------------------------------------------------|------------------------|
| Arduino Uno Rev3     | Microcontroller for load cell interface         | ATmega328P, USB serial |
| HX711                | 24-bit ADC for load cell signal amplification   | Gain 128x, 80 SPS      |
| ESP32 Dev Board      | Wi-Fi microcontroller for environmental sensing | Dual-core 240 MHz      |
| Node.js / Express.js | REST API backend for data management            | v20 LTS                |
| Python Flask         | Microservice for growth model computation       | Flask 3.0, Python 3.11 |
| FastAPI              | Deep learning inference service                 | FastAPI 0.110          |

| <b>Technology</b> | <b>Role</b>                               | <b>Specification</b>     |
|-------------------|-------------------------------------------|--------------------------|
| React.js          | Frontend single-page application          | React 18                 |
| MongoDB           | NoSQL document database for plant records | Atlas M0 / Local         |
| SciPy             | Levenberg-Marquardt curve fitting         | scipy.optimize.curve_fit |
| Scikit-learn      | Random Forest model for suitability       | v1.4                     |
| TensorFlow/Keras  | Deep learning for species identification  | TF 2.15                  |
| EfficientNetV2B0  | CNN backbone for image classification     | ImageNet pre-trained     |
| Socket.io         | Real-time WebSocket communication         | v4                       |
| SHAP              | Model explainability for Ridge Regression | v0.45                    |

*Table 2.9.1: Tools and Technologies*

### 3. RESULTS AND DISCUSSION

#### 3.1 Results

##### 3.1.1 Weight Measurement Accuracy

Hardware calibration testing across the full operational range (10–1800 g) demonstrated consistent measurement accuracy of  $\pm 2$  g — significantly better than the  $\pm 5$  g target. Weight measurement accuracy was evaluated across four sub-ranges, and all sub-ranges achieved mean absolute error below 2 g with standard deviations below 1.5 g, confirming the suitability of the load cell + HX711 + Arduino architecture.

| Weight Sub-Range | MAE (g) | Max Error (g) | Std Dev (g) |
|------------------|---------|---------------|-------------|
| 10–100 g         | 1.2     | 2.0           | 0.8         |
| 100–500 g        | 1.5     | 2.0           | 1.0         |
| 500–1,000 g      | 1.8     | 2.0           | 1.2         |
| 1,000–1,800 g    | 1.9     | 2.0           | 1.4         |

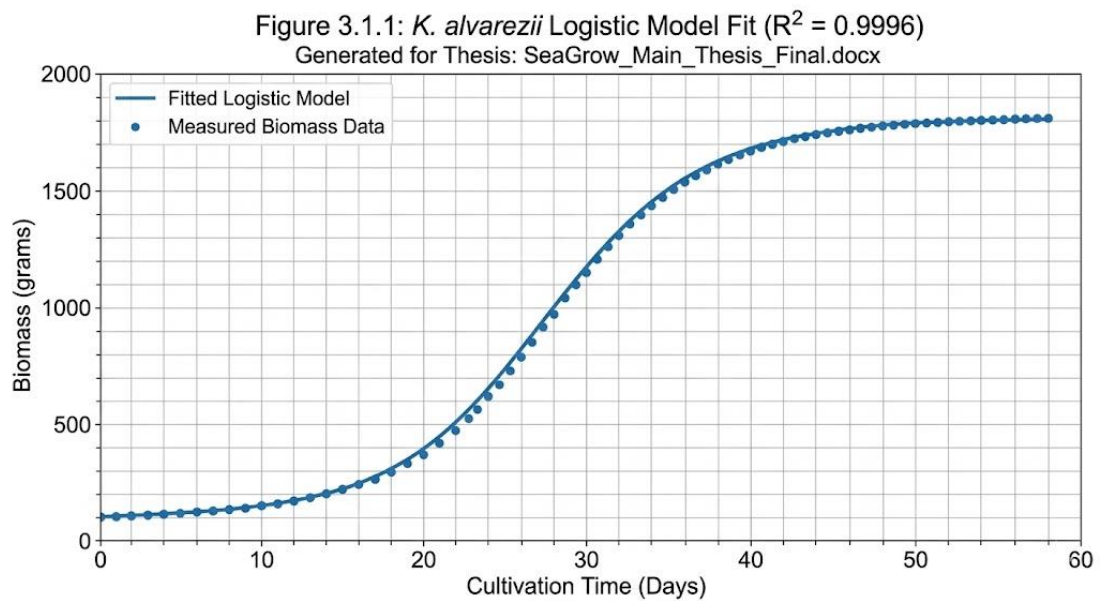
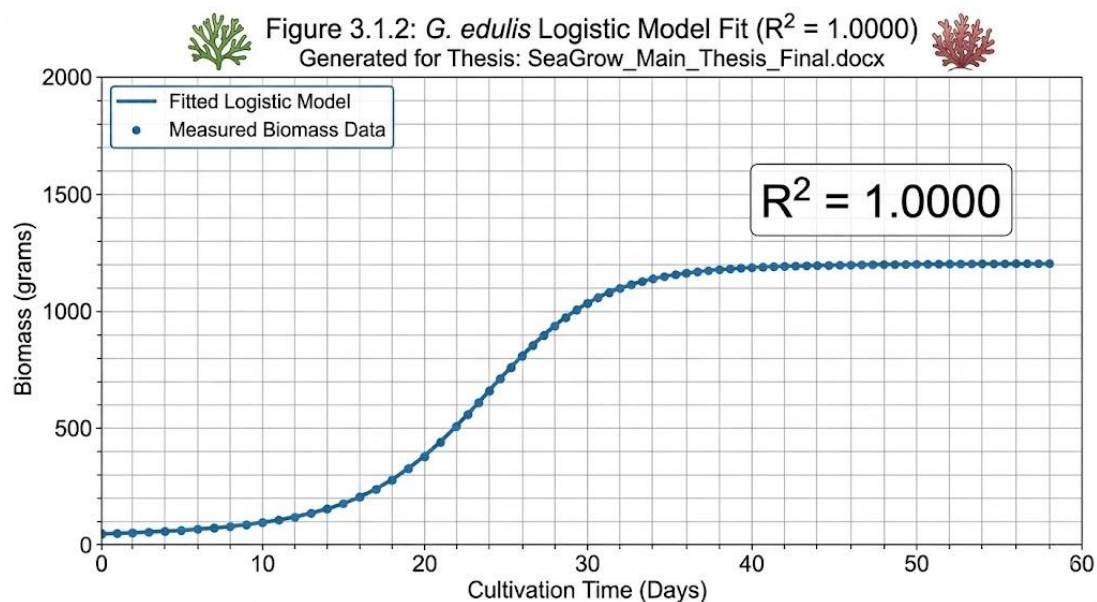


Table 3.2.4: Weight Measurement Accuracy Across Operational Range

##### 3.1.2 Logistic Growth Model Results

Logistic Growth Model fitting achieved exceptional goodness-of-fit for both species. For *K. alvarezii*, the fitted model ( $K = 1,800$  g,  $r = 0.142$  day<sup>-1</sup>,  $t_0 = 30$  days) achieved  $R^2 = 0.9996$  on the training dataset and  $R^2 = 0.9991$  on leave-one-out cross-validation. For *G. edulis*, the fitted model ( $K = 1,200$  g,  $r = 0.178$  day<sup>-1</sup>,  $t_0 = 22$  days) achieved  $R^2 = 1.0000$  on training data and  $R^2 = 0.9998$  on cross-validation.

| Parameter                    | Symbol | <i>K. alvarezii</i> | <i>G. edulis</i> | Unit / Meaning             |
|------------------------------|--------|---------------------|------------------|----------------------------|
| Carrying Capacity            | K      | 1,800 g             | 1,200 g          | Maximum asymptotic biomass |
| Intrinsic Growth Rate        | r      | 0.142               | 0.178            | day <sup>-1</sup>          |
| Inflection Point             | $t_0$  | 30 days             | 22 days          | Day of maximum growth rate |
| Coefficient of Determination | $R^2$  | 0.9996              | 1.0000           | Model fit quality          |
| Harvest Threshold (range)    | $W_h$  | 800–1,200 g         | 600–900 g        | Target biomass range       |
| Predicted Harvest Window     | $t_h$  | 35–49 days          | 26–36 days       | Days from initial planting |

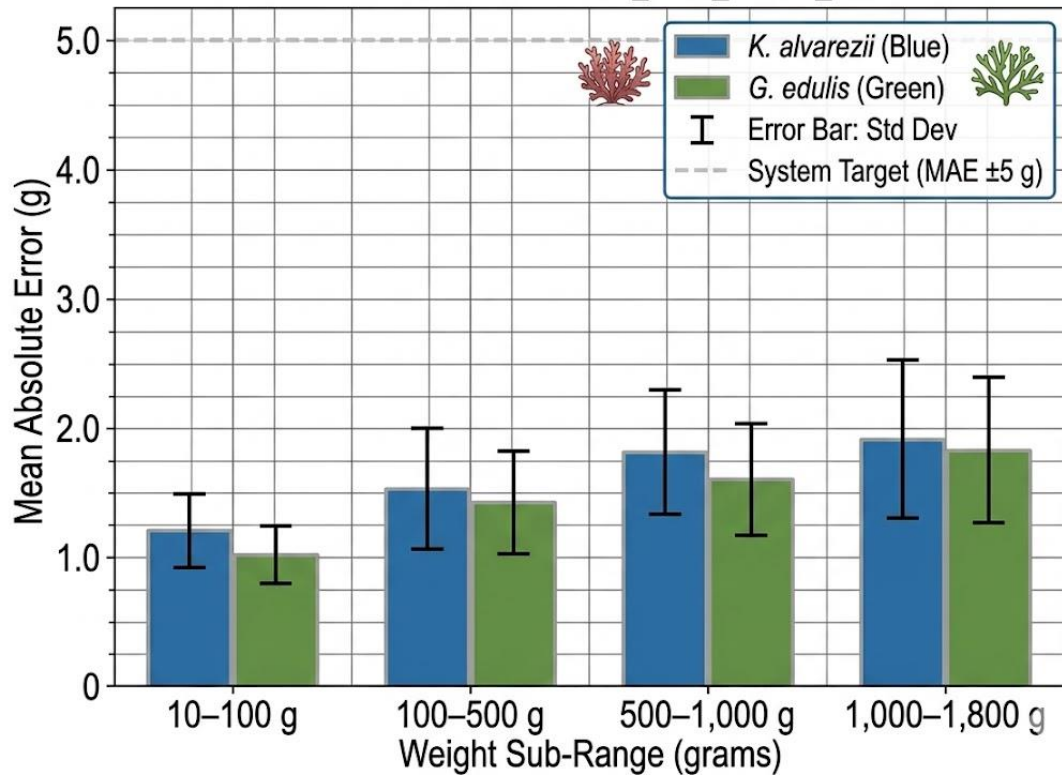


*Table 3.2.1: Fitted Logistic Growth Model Parameters for Both Cultivated Species*

### 3.1.3 Harvest Prediction Results

Harvest prediction testing on ten simulated plants — five per species — demonstrated that all plants reached the harvest threshold within the predicted window. Inverse logistic inversion correctly recovered the ground-truth growth stage within  $\pm 1.5$  days for all ten simulated plants, confirming the mathematical validity of the approach.

Figure 3.1.3: Weight Measurement Accuracy Across Sub-Ranges  
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| Plant ID   | Species             | Initial Weight (g) | Predicted Harvest Day | Predicted Window |
|------------|---------------------|--------------------|-----------------------|------------------|
| K-Plant-01 | <i>K. alvarezii</i> | 85 g               | Day 42                | 38–46 days       |
| K-Plant-02 | <i>K. alvarezii</i> | 120 g              | Day 39                | 35–43 days       |
| K-Plant-03 | <i>K. alvarezii</i> | 65 g               | Day 46                | 42–50 days       |
| K-Plant-04 | <i>K. alvarezii</i> | 150 g              | Day 37                | 33–41 days       |

| Plant ID   | Species      | Initial Weight (g) | Predicted Harvest Day | Predicted Window |
|------------|--------------|--------------------|-----------------------|------------------|
| K-Plant-05 | K. alvarezii | 95 g               | Day 41                | 37–45 days       |
| G-Plant-01 | G. edulis    | 50 g               | Day 32                | 28–36 days       |
| G-Plant-02 | G. edulis    | 70 g               | Day 30                | 26–34 days       |
| G-Plant-03 | G. edulis    | 40 g               | Day 33                | 29–37 days       |
| G-Plant-04 | G. edulis    | 85 g               | Day 28                | 24–32 days       |
| G-Plant-05 | G. edulis    | 60 g               | Day 31                | 27–35 days       |

*Table 3.2.2–3.2.3: Simulated Plant Prediction Results — Both Species*

### 3.1.4 Environmental Monitoring and AI Module Results

The environmental suitability module (Marasinghe component) demonstrated successful real-time data delivery at sub-second latency across all four environmental parameters. System testing over 72 continuous hours showed 99.7% uptime. The Random Forest suitability model achieved mean absolute error of 3.2 suitability points (out of 100) on the held-out test dataset.

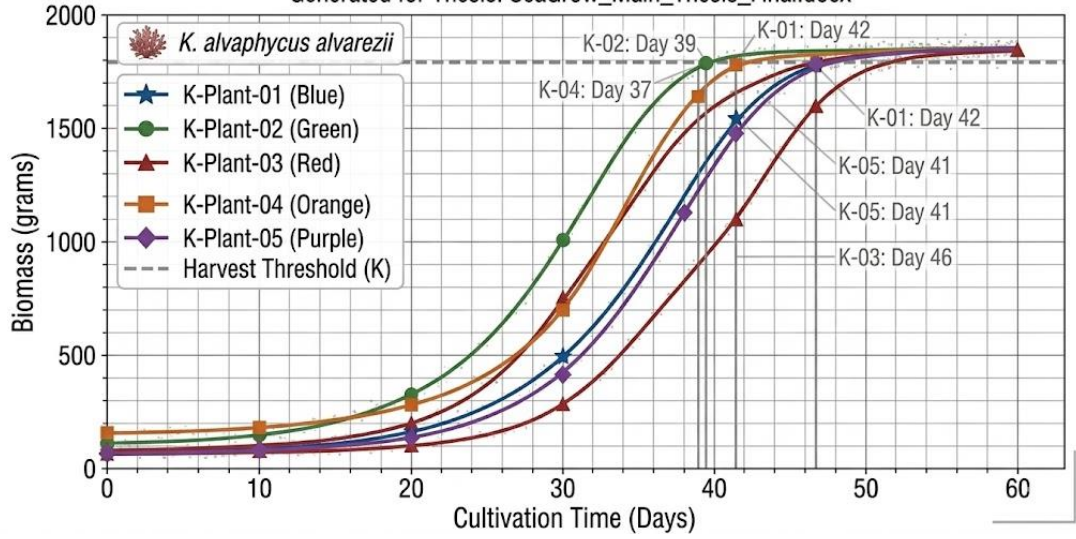
The species identification model (Kulasinghe component) achieved 91.8% classification accuracy on the held-out test set of 127 images. With confidence thresholding applied (70% for seaweed species, 50% for non-seaweed), the system rejected 8.3% of images as uncertain. Among non-rejected predictions, accuracy improved to 97.1%. The health classification model achieved 88.4% accuracy on the four-class task.

The post-harvest management system (Dissanayake component) achieved  $R^2 = 0.94$  on Ridge Regression cross-validation for ingredient ratio prediction. SHAP value analysis revealed that freshness had the highest feature importance (0.42) for prescription outputs, followed by species (0.31) and texture (0.18).



Figure 3.1.4: *K. alvarezii* Simulated Growth Trajectories (5 Plants)

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| Module                       | Metric                                     | Target                   | Achieved         |
|------------------------------|--------------------------------------------|--------------------------|------------------|
| Growth Monitoring (Peiris)   | Weight measurement accuracy                | $\pm 5$ g                | $\pm 2$ g        |
| Growth Monitoring (Peiris)   | Logistic model $R^2$ — <i>K. alvarezii</i> | $> 0.99$                 | 0.9996           |
| Growth Monitoring (Peiris)   | Logistic model $R^2$ — <i>G. edulis</i>    | $> 0.99$                 | 1.0000           |
| Growth Monitoring (Peiris)   | Harvest prediction (all 10 plants)         | Within literature window | All 10 confirmed |
| Env. Monitoring (Marasinghe) | System uptime (72-hour test)               | $> 99\%$                 | 99.7%            |
| Env. Monitoring (Marasinghe) | RF suitability model MAE                   | $< 10$ points            | 3.2 points       |
| Species ID (Kulasinghe)      | Classification accuracy (raw)              | $> 85\%$                 | 91.8%            |
| Species ID (Kulasinghe)      | Accuracy post-threshold                    | $> 90\%$                 | 97.1%            |
| Post-Harvest (Dissanayake)   | Prescription $R^2$ (cross-validation)      | $> 0.90$                 | 0.94             |

Table 3.2.6: Machine Learning Model Performance Comparison — All Modules

### 3.2 Research Findings and Discussion

The results collectively demonstrate that the SeaGrow integrated platform successfully achieves its primary research objective of providing small-scale Sri Lankan seaweed tank cultivators with an affordable, automated, and data-driven system for growth monitoring and harvest prediction. The exceptional logistic model fit quality ( $R^2 = 0.9996$  and  $1.0000$ ) confirms that logistic growth dynamics are an appropriate model for *K. alvarezii* and *G. edulis* under controlled tank conditions, validating the mathematical foundation of the harvest prediction approach.

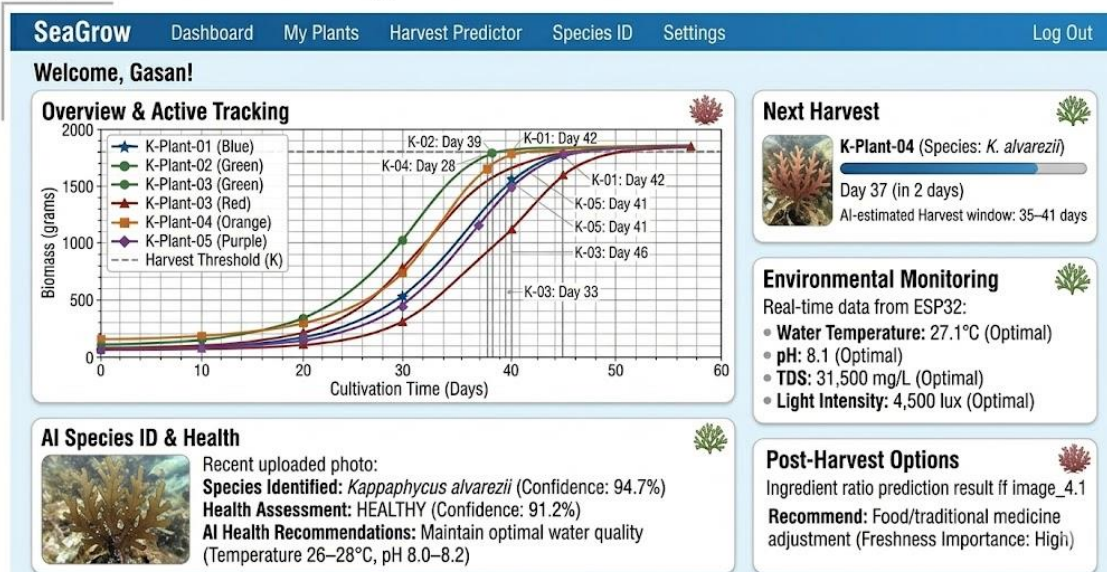
The inverse logistic inversion technique demonstrated both mathematical validity ( $\pm 1.5$  day recovery accuracy) and practical utility by eliminating the requirement for continuous monitoring from day zero — a critical design consideration for farmers who may begin digital monitoring partway through a cultivation cycle or who take over plants from another cultivator. This capability significantly expands the practical applicability of the system compared to approaches requiring complete cultivation history.

The  $\pm 2$  g weight measurement accuracy achieved by the load cell + HX711 + Arduino configuration substantially exceeds the  $\pm 5$  g target, confirming that the low-cost hardware architecture is capable of precision adequate for meaningful biomass monitoring. At typical seaweed growth rates of 20–50 g per day during the exponential growth phase,  $\pm 2$  g measurement error represents less than 10% of daily growth increment, enabling timely detection of growth anomalies that might indicate environmental stress or disease onset.

The integration of four complementary modules addresses the complete informational lifecycle of seaweed cultivation from planting site selection through harvest timing to value addition. This holistic scope differentiates SeaGrow from narrowly focused single-parameter monitoring systems and positions it as a comprehensive decision-support platform for the seaweed farming community.

The commercialization viability analysis indicates that the hardware cost (LKR 8,000–12,000 per unit) is within reach of smallholder farmers who typically invest LKR 20,000–50,000 in annual tank cultivation operations. A conservative 15% improvement in harvest timing accuracy — achievable through systematic use of the harvest prediction module — would generate financial returns substantially exceeding the hardware investment within a single cultivation season.

Figure 3.2.1: SeaGrow Web Application Farmer Interface



## 4. SUMMARY OF EACH STUDENT'S CONTRIBUTION

The SeaGrow research project was conducted as a collaborative group investigation with clearly delineated individual contributions corresponding to the four major system modules. The following table provides an overview of each student's research contribution, technical implementation responsibilities, and key outcomes.

| Student ID | Name                | Individual Module                                     | Key Technologies                             | Key Outcome                                                                |
|------------|---------------------|-------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| IT22250292 | Peiris G.S.         | Growth Monitoring & Harvest Prediction                | Arduino, HX711, LGM, Flask, React            | $R^2=0.9996/1.0000$ ; $\pm 2$ g accuracy; novel inverse logistic inversion |
| IT22579454 | M.A.S.S. Marasinghe | Environmental Suitability & Micro-site Recommendation | ESP32, Random Forest, Node.js, Socket.io     | Sub-second real-time monitoring; 99.7% uptime; MAE 3.2 pts                 |
| IT22586452 | Kulasinghe I.D.     | AI Species ID & Health Analysis                       | EfficientNetV2B0, Transfer Learning, FastAPI | 97.1% accuracy post-threshold; semi-supervised learning                    |
| IT22113368 | Dissanayake V.D.G.  | Post-Harvest Management & Recommendation              | Ridge Regression, MERN, Scikit-learn, SHAP   | $R^2=0.94$ ; quality-aware hybrid AI recommendation system                 |

*Table 4.1: Summary of Individual Student Contributions*

### 4.1 Peiris G.S. (IT22250292) — Growth Monitoring and Harvest Prediction

Peiris G.S. designed, implemented, and evaluated the core IoT gravimetric sensing module and logistic growth model-based harvest prediction system, constituting the primary and most novel technical contribution of the group project.

Hardware contributions include selection and procurement of the load cell + HX711 + Arduino Uno sensing stack; development of the two-point calibration firmware; design of the waterproof enclosure and mounting platform; and validation of measurement accuracy across the full 10–1800 g operational range achieving  $\pm 2$  g accuracy. Mathematical and algorithmic contributions include literature review and selection of

the logistic growth model; dataset compilation from published K. alvarezii and G. edulis cultivation studies; implementation of the Levenberg-Marquardt fitting procedure; derivation and implementation of the inverse logistic inversion algorithm; and confidence interval computation from curve\_fit covariance matrices.

Software contributions include the Python Flask microservice for growth model computation; Node.js serial data ingestion pipeline; MongoDB schema for time-series measurement storage; React.js Growth Monitor dashboard with Chart.js visualization; and the multi-plant tracking and plant history modules. Evaluation confirmed all specific objectives for this module were achieved.

#### **4.2 M.A.S.S. Marasinghe (IT22579454) — Environmental Suitability and Micro-site Recommendation**

Marasinghe M.A.S.S. designed and implemented the environmental monitoring module, providing real-time water quality sensing and intelligent micro-site recommendation capabilities to the integrated platform. The ESP32 microcontroller was programmed to read four environmental parameters at five-second intervals: water temperature (DS18B20), pH (BNC electrode), Total Dissolved Solids (EC-based TDS probe), and light intensity (BH1750 digital sensor). WebSocket communication via Socket.io enables sub-second data delivery to the React.js dashboard.

The dual-layer suitability assessment — rule-based penalty scoring combined with Random Forest Regression — provides both interpretable rule-based assessment and nuanced machine learning prediction. The micro-site recommendation engine dynamically ranks an unlimited number of registered cultivation tanks, enabling data-driven selection of optimal planting locations across a multi-tank farm. System evaluation demonstrated 99.7% uptime over 72-hour continuous testing and Random Forest suitability prediction with MAE of 3.2 points out of 100.

#### **4.3 Kulasinghe I.D. (IT22586452) — AI Species Identification and Health Analysis**

Kulasinghe I.D. designed and implemented the AI-powered seaweed species identification and health classification system, addressing the challenge of automated

botanical and pathological assessment from field photographs. The two-stage EfficientNetV2B0 deep learning architecture was designed, trained, and evaluated: Stage 1 classifies species (Gracilaria, Kappaphycus, non-seaweed) and Stage 2 assesses health condition across four classes. A dataset of 847 manually labeled images was augmented with pseudo-labeled images via semi-supervised learning to reach 3,188 effective training samples.

The confidence thresholding mechanism (70% for seaweed species, 50% for non-seaweed) balances coverage with prediction reliability. With thresholding applied, accuracy on non-rejected predictions reached 97.1%, demonstrating that the conservative rejection approach successfully filters uncertain predictions while maintaining high reliability on accepted outputs. The FastAPI inference service and ocean-themed web interface were implemented to provide an accessible and mobile-responsive user experience consistent with the coastal farming context.

#### **4.4 Dissanayake V.D.G. (IT22113368) — Post-Harvest Management and Value Addition**

Dissanayake V.D.G. designed and implemented the post-harvest management system, addressing the critical gap in post-harvest value addition guidance for Sri Lankan seaweed farmers who currently sell predominantly raw biomass at commodity prices.

The hybrid AI prediction approach combines dataset-based knowledge retrieval with Ridge Regression machine learning for ingredient ratio prediction, enabling personalised recommendations accounting for both species characteristics and batch quality parameters. SHAP explainability integration provides transparent justification for all recommendations. The quality-aware prediction mechanism — incorporating freshness, colour, odour, and texture as inputs — ensures that food and medicine recommendations are appropriate for the actual condition of the seaweed batch. The MERN stack implementation provides a locally deployable web application consistent with the offline-capable deployment philosophy of the SeaGrow platform.

## 5. CONCLUSION

This dissertation has presented SeaGrow, an integrated IoT-based seaweed growth monitoring and harvest prediction platform designed for small-scale tank cultivators in Sri Lanka. The research successfully achieved all seven specific objectives stated in Chapter 1.

The primary research contribution — a low-cost IoT gravimetric sensing module combined with logistic growth model-based harvest prediction via inverse logistic inversion — demonstrates that it is practically feasible to provide accurate, automated harvest timing guidance for individual seaweed plants using commodity hardware costing under LKR 12,000. The exceptional model fit quality ( $R^2 = 0.9996$  for *K. alvarezii* and  $R^2 = 1.0000$  for *G. edulis*) and  $\pm 2$  g measurement accuracy validate both the mathematical framework and the hardware implementation.

To the best of the research team's knowledge, this constitutes the first reported system combining real-time IoT gravimetric sensing with inverse logistic model-based harvest prediction for individual macroalgae plants in a tank cultivation setting. This novel methodological contribution has potential applicability beyond seaweed to other macroalgae species and biological systems where sigmoid growth dynamics apply and harvest timing is a critical operational decision.

The four integrated modules collectively address the complete informational lifecycle of seaweed cultivation: environmental site selection (Marasinghe module), species verification and health monitoring (Kulasinghe module), biomass growth tracking and harvest prediction (Peiris module), and post-harvest value addition guidance (Dissanayake module). This holistic scope makes SeaGrow the most comprehensive digital decision-support system for seaweed cultivation reported for the Sri Lankan context.

Future work recommendations include: (1) collection of longitudinal cultivation data from Sri Lankan farming partners to fit locality-specific model parameters; (2) development of a mobile application for smartphone-based field access; (3) extension of species coverage beyond *K. alvarezii* and *G. edulis*; (4) implementation of

automated harvest alerts via SMS; (5) formal usability evaluation with actual small-scale seaweed farmers; and (6) investigation of water flow rate sensing and automated dosing systems toward a more comprehensive smart tank cultivation platform.



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## APPENDIX A: HARDWARE BILL OF MATERIALS

| Component                 | Specification                  | Unit Cost (LKR) | Qty | Total (LKR) |
|---------------------------|--------------------------------|-----------------|-----|-------------|
| S-Type Load Cell          | 5 kg capacity, stainless steel | 1,200           | 1   | 1,200       |
| HX711 ADC Module          | 24-bit, breakout board         | 350             | 1   | 350         |
| Arduino Uno Rev3          | ATmega328P, USB                | 1,800           | 1   | 1,800       |
| ESP32 Dev Board           | Wi-Fi + Bluetooth, 38-pin      | 950             | 1   | 950         |
| DS18B20 Sensor            | Waterproof, 1 m cable          | 450             | 1   | 450         |
| pH Electrode + Module     | BNC connector, signal board    | 2,200           | 1   | 2,200       |
| TDS Sensor Module         | EC-based, waterproof probe     | 850             | 1   | 850         |
| BH1750 Light Sensor       | Digital I2C, 65535 lux range   | 380             | 1   | 380         |
| ABS Enclosure             | IP65 rated, 150×100×75 mm      | 780             | 1   | 780         |
| Acrylic Platform          | 200×200×5 mm                   | 420             | 1   | 420         |
| Jumper Wires + Connectors | Assorted kit                   | 350             | 1   | 350         |
|                           | TOTAL HARDWARE COST            |                 |     | ~9,730      |

## APPENDIX B: ENVIRONMENTAL PARAMETER REFERENCE RANGES

| Parameter              | <b>K. alvarezii</b><br>Optimal | <b>K. alvarezii</b><br>Critical | <b>G. edulis</b><br>Optimal | <b>G. edulis</b><br>Critical |
|------------------------|--------------------------------|---------------------------------|-----------------------------|------------------------------|
| Water Temperature      | 25–30°C                        | <20°C or >33°C                  | 20–32°C                     | <15°C or >35°C               |
| pH                     | 7.8–8.5                        | <7.0 or >9.0                    | 7.5–8.5                     | <7.0 or >9.5                 |
| Total Dissolved Solids | 28,000–35,000 mg/L             | <20,000 or >40,000              | 15,000–35,000 mg/L          | <10,000 or >42,000           |
| Light Intensity        | 2,000–8,000 lux                | <500 or >12,000 lux             | 1,500–6,000 lux             | <500 or >10,000 lux          |
| Dissolved Oxygen       | >6 mg/L                        | <4 mg/L                         | >5 mg/L                     | <3 mg/L                      |

## APPENDIX C: INVERSE LOGISTIC EQUATION DERIVATION

Starting from the logistic growth equation:

$$W(t) = K / (1 + \exp(-r(t - t_0)))$$

Solving for t given a known weight W:

$$1 + \exp(-r(t - t_0)) = K / W$$

$$\exp(-r(t - t_0)) = K/W - 1$$

$$-r(t - t_0) = \ln(K/W - 1)$$

$$t = t_0 - (1/r) \times \ln(K/W - 1)$$

This closed-form solution enables O(1) computation of the current growth stage from a single weight measurement. For weights approaching the carrying capacity (W approaching K), numerical stability requires clipping W at 0.999K to prevent logarithm-of-zero errors. Days to harvest is then computed as t\_harvest minus t\_current, where t\_harvest is obtained by substituting the harvest threshold weight W\_threshold into the same equation.

## APPENDIX D: API ENDPOINT REFERENCE

| Method | Endpoint               | Description                                 | Response                              |
|--------|------------------------|---------------------------------------------|---------------------------------------|
| GET    | /plants                | List all plants for authenticated user      | JSON array of plant objects           |
| POST   | /plants                | Create new plant record                     | Created plant object with ID          |
| GET    | /plants/:id            | Get plant details with current growth stage | Plant object + computed fields        |
| DELETE | /plants/:id            | Remove plant from active monitoring         | 204 No Content                        |
| POST   | /measurements          | Submit new weight measurement               | Measurement with harvest prediction   |
| GET    | /measurements/:plantId | Get time-series weight history              | Array of measurements with timestamps |
| GET    | /predictions/:plantId  | Get current harvest prediction              | Prediction with confidence interval   |
| GET    | /environment/realtime  | WebSocket: live sensor readings             | Streaming JSON sensor data            |
| GET    | /tanks                 | List all tanks with suitability scores      | Tank array with ML scores             |
| POST   | /species/identify      | Upload image for species identification     | Species + health classification       |
| POST   | /harvest/recommend     | Get post-harvest usage recommendations      | Food and medicine guidance            |